MACHINE LEARNING CANVAS

Date: 16/10/2026 Iteration: 1 (Etapa 1).

Designed for: MiBici (operador, rebalanceo y planeacion urbana) Designed by: Equipo 49 (TC5061)

PREDICTION TASK

What is the type of task?
Which entity are predictions made on?
What are the possible outcomes to predict?
When are outcomes observed?

P1 Regresion: viajes que salen de cada estacion en cada hora (conteo, entero >= 0).
P2 Clasificacion: riesgo 0/1 de que la estacion quede vacia o llena en las proximas 2 h.
P3 Clustering: grupos de estaciones por patron de uso (k entre 3 y 8).
Entidad: estacion (484 en catalogo; 469 con salidas en 12 meses).
Cuando se observa: P1 y P2 al cierre de cada hora; P3 con el periodo de entrenamiento.

DECISIONS

How are predictions turned into actionable recommendations or decisions for the end-user? (Mention parameters of the process / application for this.)

P1: pronostico diario a las 00:00 para las 24 h. Reparte bicicletas y reservas por estacion antes del pico de 08:00.
P2: lista de estaciones en riesgo, ordenada. El equipo visita las k mas criticas por hora (k por confirmar).
P3: grupos guian donde expandir y que tipo de anclajes poner.
Umbral P2 elegido en validacion con precision minima de 0.5.

VALUE PROPOSITION

Who is the end beneficiary, and what specific pain points are addressed?
How will the ML solution integrate with their workflow, and through which user interfaces?

Beneficiarios: operador de MiBici (planeacion de flota), equipo de rebalanceo y planeacion urbana.
Dolores: estaciones vacias o llenas en hora pico; reservas mal ubicadas; expansion sin datos de patron.
Integracion: reporte diario en tabla y mapa (CSV o panel). Sin interfaz propia en esta etapa.

DATA COLLECTION

How is the initial set of entities and outcomes sourced (e.g., database extracts, API pulls, manual labeling)?
What strategies are in place to update data continuously while controlling cost and maintaining freshness?

Carga mensual de los CSV del portal: 12 meses (oct-2025 a sep-2026), 4,497,877 viajes antes de limpieza.
Ingesta reproducible (dataset.py) con manifiesto SHA-256 por archivo.
Frescura: un mes nuevo al mes. Costo nulo (archivos abiertos de 27 a 42 MB).
Etiquetas derivadas de los viajes; no hay etiquetado manual.

DATA SOURCES

Where can we get data on entities and observed outcomes? (Mention internal and external database tables or API methods.)

Viajes: mibici.net/es/datos-abiertos (CSV mensual). Campos: Inicio, Fin, Origen_Id, Destino_Id, Viaje_Id, Usuario_Id, Genero, Año de nacimiento.
Estaciones: nomenclatura_2026_09.csv (id, nombre, lat, lon, capacidad).
Sin inventario de bicicletas por estacion ni eventos de rebalanceo: limita P2.
Externos previstos: festivos de Mexico. Clima fuera de esta etapa.

IMPACT SIMULATION

What are the cost/gain values for (in)correct decisions?
Which data is used to simulate pre-deployment impact?
What are the criteria for deployment?
Are there fairness constraints?

Costo y beneficio: no hay datos financieros de MiBici. Por confirmar con el operador: costo de visita vs. viaje perdido.
Datos de simulacion: validacion (jul-ago 2026) y prueba (sep 2026, 490,056 viajes).
Criterio: P2 supera la persistencia en PR-AUC; P1 supera el estacional-ingenuo en WAPE.
Equidad: revisar que el error no se concentre en clusters de baja demanda.

MAKING PREDICTIONS

Are predictions made in batch or in real time?
How frequently?
How much time is available for this (including featurization and decisions)?
Which computational resources are used?

Por lotes. P1: una vez al dia. P2: cada hora. P3: mensual o trimestral.
Tiempo: menos de 10 min por corrida (unas 470 estaciones).
Recursos: CPU de equipo personal o servidor pequeño. Sin GPU.

BUILDING MODELS

How many models are needed in production?
When should they be updated?
How much time is available for this (including featurization and analysis)?
Which computation resources are used?

Tres modelos en produccion, uno por pregunta.
Reentrenar cada mes con ventana movil de 12 meses; P3 cada trimestre o al cambiar la red.
Tiempo: pocos minutos por corrida con scikit-learn (CPU).
Registro y comparacion con MLflow Tracking.

FEATURES

What representations are used for entities at prediction time?
What aggregations or transformations are applied to raw data sources?

Calendario: hora, dia de la semana, mes, festivo y fin de semana.
Rezagos de 24, 48 y 168 h; media de 4 semanas para el mismo dia y hora.
Capacidad y cluster de P3 (ajustado solo con entrenamiento).
P2: flujo neto acumulado hasta t; llegadas y salidas de la ultima hora y de las ultimas dos.
Sin fuga: solo informacion anterior al momento de prediccion. No se usan Usuario_Id, genero ni edad.

MONITORING

Which metrics and KPIs are used to track the ML solution's impact once deployed, both for end-users and for the business?
How often should they be reviewed?

Negocio: viajes perdidos por estacion vacia o llena; visitas en riesgo.
Modelo: WAPE, MAE (P1); PR-AUC (P2); silueta (P3). Alerta si empeora 10%.

Version 1.2. Created by Louis Dorard, Ph.D. Licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
Please keep this mention and the link to ownml.co when sharing.

What your canvas is for

A filled canvas does 4 jobs, and each one needs certain boxes filled in with care.

- **Deciding what's worth building.** Put each idea on its own canvas and you can compare the ideas on the same terms: what each could gain in a year, and whether the outcomes a model would learn from were ever recorded. That takes dollar amounts in the Impact Simulation box, and a named person in the Value Proposition box who acts on each prediction.
- **Agreeing on what gets built.** Some of the choices that set a system's value look technical, but they're business decisions. What counts as a lost customer: 60 days without buying, or 90? How far ahead do you predict? How many flagged cases can your team act on? On the canvas, business, data and engineering make these choices together, instead of leaving them to whoever writes the code.
- **Knowing when the model is good enough.** The Impact Simulation box sets the performance the model has to reach to pay off. Test the model on enough of the cases you're predicting: 35,000 machine-hours of data can hold just 3 failures, too few to measure anything.
- **Learning after launch.** If acting on a prediction changes the outcome, keep a small share of flagged cases untouched and record that share in the Decisions box, so you can see what your action changed.

Whatever you can't answer is the work left before anyone writes code.

"The Canvas allowed us to structure and focus our delivery sprints. The ML system went live in June 2020 and still is."
— Christian Schiller, Senior IT & Enterprise Architect, Deutsche Telekom

Book a project review

Louis Dorard created the canvas in 2015. Book a free 30-minute call, then send him your canvas draft, or a few lines about the project if you haven't started one. He prepares the call from what you send, and you'll leave knowing where the project is likely to break, and what to check before anyone builds it.

Book a 30-minute call →

Reference the Machine Learning Canvas

If you need to reference the Canvas formally (e.g. in documentation or publications), you may use:

```
@misc{mlc,
  title  = "Machine Learning Canvas",
  author = "Louis Dorard (OWNML)",
  url    = "https://www.ownml.co/",
  year   = "2015"
}
```

"MLC played an essential role in a positive project outcome. It is now being used throughout the organization."
— Daitan, now part of Coforge, a global IT services firm